(vi) $F'(x)=\frac{d}{dx}F(x)=f(x)\Rightarrow dF(x)=f(x)dx$

(vii) $dF(x)$ is known as probability differential of X.

(viii) $P(a\le x\le b)=\int_a^b f(x)dx=\int_{-\infty}^b f(x)dx-\int_{-\infty}^a f(x)dx$

$=P(X\le b)-P(X\le a)$

$=F(b)-F(a)$

- Mathematical Expectation

  The expected value is a weighted average of the values of a random variable may assume.

- Discrete random variable with probability mass function (p.m.f.)

  $E(X)=\sum_x x\,p(x)$

- Continuous random variable with probability density function

  $E(X)=\int_{-\infty}^{\infty} x\,f(x)\,dx$

- The mean or expected value of X, denoted by φ_X or E(X).
- The variance is a weighted average of the squared deviations of a random variable from its mean.
- $Var(X)=\sum\left[x-E(X)\right]^2 p(x)$

  if X is discrete random variable with probability mass function $p(x)$.

- $Var(X)=\int_{-\infty}^{\infty}\left[X-E(X)\right]^2 f_X(x)\,dx$

  if X is continuous random variable with probability density function $f_X(x)$.

- Expected value of $[X-E(X)]^2$ is called the variance of the random variable.

  i.e., $Var(X)=E\left[X-E(X)\right]^2=E\left(X^2\right)-\left[E(X)\right]^2$

  where $E\left(X^2\right)=\begin{cases}\sum_x x^2\,p(x), & \text{if } X \text{ is Discrete Random Variable}\\ \int_{-\infty}^{\infty} x^2 f(x)\,dx, & \text{if } X \text{ is Continuous Random Variable}\end{cases}$

- If X is a random variable, the standard deviation of X, denoted by σ_X, is defined as $+\sqrt{Var[X]}$.
- The variance of X, denoted by $\sigma_X{}^2$ or $Var(X)$ or $V(X)$.
- Properties of Mathematical expectation

  (i) $E(a)=a$, where 'a' is a constant

  (ii) $E(aX)=aE(X)$

  (iii) $E(aX+b)=aE(X)+b$, where 'a' and 'b' are constants.

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